Resting HRV, Frontal Midline Theta, and Working Memory Updating

A Psychophysiological Research Proposal

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ABSTRACT

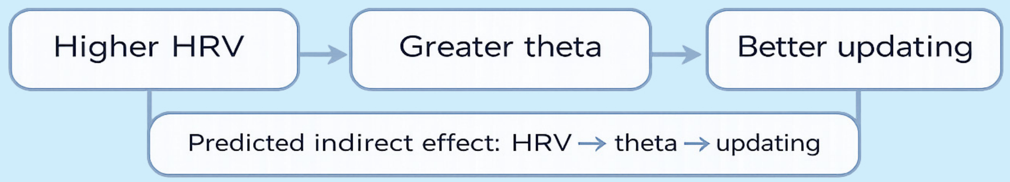
- This study tests whether resting vagally mediated HRV predicts working memory updating.
- It also tests whether frontal midline theta statistically mediates this relationship.
- Participants will complete a 5-minute resting ECG and an EEG n-back task.
- Outcomes: accuracy, d', and reaction time.
- The study links autonomic, neural, and behavioural measures of executive control.

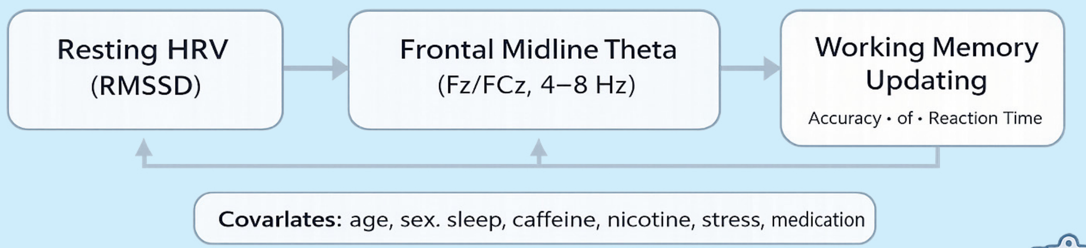
INTRODUCTION

- Working memory updating = replacing outdated information with task-relevant material.
- It is a core part of executive control (Andrewes, 2015).
- Resting HRV has been linked to self-regulation and executive functioning, although findings are mixed (Magnon et al., 2022; Nicolini et al., 2024).
- Frontal midline theta is a reliable EEG marker of cognitive control (Cavanagh & Frank, 2014).
- Research question: Does resting HRV predict working memory updating, and is this relationship mediated by frontal midline theta?
- Hypotheses: Higher HRV and greater theta will predict better updating; theta will mediate the HRV-updating relationship.

RESEARCH DESIGN & METHOD

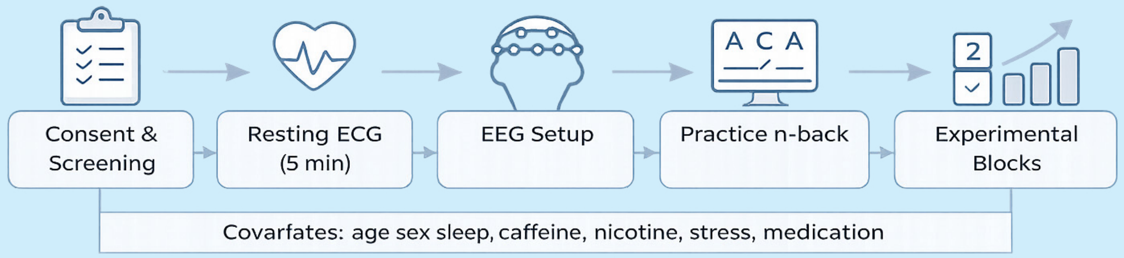
- Design: Correlational, cross-sectional, single-session study.
- Participants: healthy adults aged 18–35.
- Target sample: 115 recruited to retain an analysable sample of approximately 103 after artefact rejection.
- HRV: 5-minute ECG baseline; RMSSD.
- Theta: EEG at Fz/FCz, 4–8 Hz.
- Performance: accuracy, d', reaction time.
- Procedure: Consent » ECG » EEG setup » practice n-back » experimental blocks » analysis.





LIMITATIONS

- The design is cross-sectional, so causal conclusions cannot be drawn.
- The n-back task is not process-pure.
- HRV is sensitive to sleep, stress, caffeine, nicotine, and medication.
- EEG has limited spatial precision.
- EEG preparation may cause mild discomfort, and participants may stop at any time.
- Physiological data will be anonymised and stored securely.



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